A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Ionising radiation - Set 1

Mark scheme

AQA spec 3.4.3.2.14; Pollution; textbook pp. 295-304

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	1 mark for the point below. **1** The time taken for half of the atoms of a radioactive isotope to decay (activity halves) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **half-life** → the time for the activity, or the amount, to fall by half **Do not credit** • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues	1

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 32 / 8 = 4 half-lives **1** 6400 / 2^4 = 400 Bq **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (Bq) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
01.3	**1 mark each** 1 mark for each of the 3 points below. **1** Short half-life isotopes (I-131) release all their radiation quickly: dangerous but only briefly so short-term precautions (potassium iodide tablets, evacuation) are sufficient **1** Very long half-life isotopes (U-238) emit only small amounts of radiation in any period so the level of danger is low **1** Decades-long half-lives combine a high rate of radiation release with persistence for most or all of a human lifetime, so short-term protection is inadequate (land contaminated for generations) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **half-life** → the time for the activity, or the amount, to fall by half • **persistence** → does not break down / stays in the environment for a long time **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues	3
01.4	**1 mark each** 1 mark for each of the 3 points below. **1** Alpha particles have very low penetrating power and are absorbed by clothing / dead skin so external sources cause little exposure **1** Inside the body the alpha particles are absorbed in a very small mass of tissue so damage is concentrated (high RBE - weighting factor 20) **1** Long half-life means it continues to emit for the rest of the person's life / cannot be excreted quickly (contamination not just exposure) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **half-life** → the time for the activity, or the amount, to fall by half **Do not credit** • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues	3

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Question	Marking guidance	Marks
01.5	1 mark for the point below. **1** A previously stable nucleus that becomes radioactive after absorbing a neutron (eg in reactor structures), later decaying to emit ionising radiation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ionising radiation** → radiation with enough energy to remove electrons and form ions	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 299-301; AQA spec 3.4.3.2.14.	
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** Radiation absorbed by tissue produces free radicals (ions with unpaired electrons), mostly from water, which are highly reactive **1** Free radicals damage DNA / chromosomes in the nucleus causing mutations (cancer, inherited effects) - rapidly dividing cells (skin, gut, bone marrow) most affected **1** Damage to other cell parts can stop the cell functioning or kill it – low damage may be repaired by the intact nucleus **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ionising radiation** → radiation with enough energy to remove electrons and form ions	3
02.2	**1 mark each** 1 mark for each of the 2 points below. **1 Acute:** effects appearing quickly after a large dose over a short time (radiation sickness, damage to bone marrow / immune system / gut, haemorrhage, rapid death) **1 Chronic:** effects appearing slowly, in proportion to the dose received over a long period (cancers, cataracts) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.3	**1 mark each** 1 mark for each of the 2 points below. **1 Exposure:** absorption of ionising radiation from a source the person is close enough to **1 Contamination:** physically carrying radioactive material on or in the body, so exposure continues wherever the person goes – sealed sources can expose but not contaminate **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ionising radiation** → radiation with enough energy to remove electrons and form ions	2

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Question	Marking guidance	Marks
02.4	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Sieverts = grays x radiation weighting factor (20 for alpha) **1** 0.002 x 20 = 0.04 Sv (40 mSv) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (Sv) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
02.5	1 mark for the point below. **1** The activity of a source (number of decays per second) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **becquerel** → one radioactive decay per second	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 301, 303; AQA spec 3.4.3.2.14.	
03.1	**1 mark each** 1 mark for each of the 2 points below. **1** Doubling the distance from 1 m to 2 m reduces the dose rate from 400 to 100 (one quarter) **1** Distance x 10 (1 to 10 m) reduces it by 100 times (400 to 4) – dose is proportional to 1/distance^2 **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.2	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 400 / 20^2 = 1 μSv h^{-1} **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (μSv h^{-1}) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1

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Question	Marking guidance	Marks
03.3	**1 mark each** 1 mark for each of the 4 points below. **1** Sealed container prevents contact / contamination so exposure stops when the worker moves away – container may absorb alpha completely – remote handling **1** Barriers (lead, concrete, thick steel flask walls) absorb radiation before it reaches the worker – thickness depends on radiation type **1** Dose received is proportional to time near the source so work is organised to minimise time close to it (job rotation) **1** Dosemeters / film badges record exposure – air monitors detect alpha-emitting particles – contamination monitors on leaving check that no material is carried out **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** Stable iodine saturates the thyroid **1** So radioactive iodine-131 released is not taken up / concentrated in the thyroid gland **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	2
03.5	1 mark for the point below. **1** Exposure should be As Low As Reasonably Achievable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 302-303; AQA spec 3.4.3.2.14.	
04.1	**1 mark each** 1 mark for each of the 3 points below. **1** The underlying granite contains uranium whose decay produces the radioactive gas radon (an alpha emitter) **1** Radon travels up cracks in the rock and through foundations into buildings **1** It becomes concentrated inside enclosed / poorly ventilated buildings – control by ventilating foundations / sumps **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
04.2	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 16 + 1.5 = 17.5% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
04.3	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Where they live (geology / radon, altitude) **1** Occupation (radiographer, nuclear worker, aircrew, miner) **1** Medical history (number of X-rays / CT scans) **1** Lifestyle / diet (eg shellfish near discharges) – smoking **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	2
04.4	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** X-rays carry a small risk of mutation / cancer but the benefit (diagnosis, avoiding invasive surgery) may outweigh it – risks can be reduced (shielding, dose limits) **1** Symptoms of chronic low-dose exposure take a long time to develop **1** The same symptoms (cancers) have other causes so effects are hard to attribute **1** Accurate data on human exposure / doses is often unavailable (especially after accidents) **1** People who benefit and people who take the risk may be different (eg uranium miners vs electricity consumers) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ionising radiation** → radiation with enough energy to remove electrons and form ions • **Accurate** → close to the true value	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 297-299; AQA spec 3.4.3.2.14.	

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Question	Marking guidance	Marks
05.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 High-level:** used fuel rods / fission products separated in reprocessing – vitrification in molten glass in sealed stainless steel containers, surrounded by concrete, air-cooled to remove decay heat (Sellafield) **2 Intermediate:** fuel cladding / metal tubes around fuel rods, filters from reprocessing – mixed with cement (encapsulated) in stainless steel drums, stored in concrete vaults **2 Low-level solid:** contaminated clothing, equipment – sealed in plastic bags inside steel drums / containers in a concrete-lined, monitored landfill (Drigg) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **vitrification** → sealing waste into glass **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	6
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Radioactive decay of the concentrated fission products generates heat which could damage containers **1** Isotopes with long half-lives (plutonium 24 400 years) remain hazardous for thousands of years **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.3	**1 mark each** 1 mark for each of the 2 points below. **1 Liquids:** filtered and passed through ion-exchange (sands / resins) to remove radionuclides, then discharged (monitored by CPA / CGM) **1 Gases:** filtered (HEPA / charcoal filters) to remove particles and iodine before release **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 303; AQA spec 3.4.3.2.14.	

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Question	Marking guidance	Marks
06.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Uses / benefits:** nuclear electricity (low carbon, high energy density), medicine (X-rays, CT, radiotherapy, tracers, sterilising equipment), industry (thickness gauges, polymer cross-linking, testing turbines, oil exploration), agriculture (food irradiation, sterile male technique, mutation breeding), research tracers, smoke detectors, ship / submarine propulsion • **Risks:** free radical production, DNA damage, mutation, cancer – acute and chronic, somatic and gonadic effects – contamination vs exposure – activation products – long-lived wastes – accidents (Chernobyl, Fukushima) – weapons fallout • **Factors affecting risk:** type of radiation (alpha, beta, gamma, neutrons - penetration, RBE / weighting factors), half-life, dose (Gy, Sv), route of exposure • Risk:benefit analysis and why it is difficult (latency, alternative causes, poor exposure data, risks and benefits to different people, offsets such as avoided surgery) • **Principles:** ALARA, BATNEEC – closed sources, remote handling, absorbers / shielding, distance (inverse square law), time, protective clothing, decontamination, potassium iodide • **Monitoring:** dosemeters, film badges, air and contamination monitors – environmental monitoring via CPA and CGM – waste categories and disposal (vitrification, encapsulation, Drigg) • **Evaluation / judgement:** benefits generally outweigh risks when controls are applied – public perception vs actual risk (background radon 50% of dose vs <1% from nuclear industry)	9

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	Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **ionising radiation** → radiation with enough energy to remove electrons and form ions • **half-life** → the time for the activity, or the amount, to fall by half • **vitrification** → sealing waste into glass **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues	
	Model answer - Level 3 (8 marks)	
	Ionising radiation is used because it does things nothing else can: nuclear fission generates low-carbon electricity from fuel of extraordinary energy density; medicine relies on X-rays and CT scans, radiotherapy, tracers and gamma sterilisation of equipment; industry uses thickness gauges, cross-links polymers, tests turbine blades and explores for oil; and agriculture uses irradiation to preserve food and sterilise insect pests. The risks arise because radiation produces free radicals that damage DNA, causing mutation and cancer, with acute and chronic, somatic and gonadic effects; contamination is worse than exposure because the source stays with the person; and activation products, long-lived wastes, accidents at Chernobyl and Fukushima and weapons fallout add environmental risks.	
	Risk depends on the type of radiation - alpha is barely penetrating but has a weighting factor of 20 - on half-life, on dose in grays and sieverts and on the route of exposure. Weighing risks against benefits is difficult because cancers appear years later with many other possible causes, exposure data are poor, the people who benefit are often not those at risk, and benefits such as avoided surgery are hard to quantify.	
	Control follows ALARA and BATNEEC: closed sources, remote handling, shielding, distance under the inverse-square law, limiting time, protective clothing, decontamination and potassium iodide against iodine-131. Workers wear dosemeters and film badges and pass contamination monitors; discharges are managed by Critical Pathway Analysis and Critical Group Monitoring; wastes are vitrified, encapsulated or stored at licensed sites. With these controls the benefits generally outweigh the risks - natural radon supplies about half the average dose and the nuclear industry under one per cent - although public perception of risk remains far higher than the measured risk.	
	Weaker answer - Level 2 (5 marks)	
	Ionising radiation is used in nuclear power stations to make electricity, in hospitals for X-rays and treating cancer, and in industry. The risks are that radiation damages DNA and causes cancer, and accidents like Chernobyl release radiation into the environment. Nuclear waste stays dangerous for a long time. Workers are protected by shielding, keeping their distance and wearing film badges to measure their dose. The public is protected by monitoring around nuclear sites. Overall the benefits are worth the risks if it is controlled properly.	
	What makes the difference	

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Question	Marking guidance	Marks
	• Level 2 covers uses, risks, controls and a judgement, but each briefly; Level 3 adds the factors that determine risk (type, half-life, dose units, route) and why risk-benefit analysis is difficult.	
	• Level 3 names the principles (ALARA, BATNEEC, inverse square law, CPA / CGM) rather than describing practice loosely.	
	• The judgement in Level 3 is supported with evidence (radon vs nuclear industry doses); Level 2's is an opinion.	
06.2	1 mark for the point below. **1** Consumers of the electricity may live in a different country / region from the uranium miners or the communities near waste stores who bear the risks **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 295-304; AQA spec 3.4.3.2.14.	

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